

30 Days SQL Micro Course

Day 2 – Types of Commands in SQL

5 types Commands in SQL



5 Types of SQL Commands

SQL commands are divided into 5 categories:

Type	Full Form	Purpose
DDL	Data Definition Language	Defines database structure
DML	Data Manipulation Language	Works with data
DQL	Data Query Language	Retrieves data
TCL	Transaction Control Language	Manages transactions
DCL	Data Control Language	Controls permissions

1. DDL (Data Definition Language)

Used to create, modify, and delete database objects like tables.

Command	Purpose
CREATE	Create a table/database
ALTER	Modify a table
DROP	Delete a table
TRUNCATE	Remove all rows from a table

Examples

```
CREATE TABLE employee (
  id INT,
  name VARCHAR(50)
);
```

```
ALTER TABLE employee
ADD age INT;
```

```
DROP TABLE employee;
```

```
TRUNCATE TABLE employee;
```

Easy Memory Trick

E DDL = Design the Database

2. DML (Data Manipulation Language)

Used to add, update, and delete data inside tables.

Command	Purpose
INSERT	Add data
UPDATE	Modify data
DELETE	Remove data

Examples

```
INSERT INTO employee  
VALUES (1,'Rahul');
```

```
UPDATE employee  
SET name='Priya'  
WHERE id=1;
```

```
DELETE FROM employee  
WHERE id=1;
```

Easy Memory Trick

 DML = Modify Data

3. DQL (Data Query Language)

Used to view/retrieve data.

Command	Purpose
SELECT	Fetch data

Example

```
SELECT * FROM employee;
```

Easy Memory Trick

 DQL = Query (See) Data

4. TCL (Transaction Control Language)

Used to manage transactions.

Command	Purpose
BEGIN	Start transaction
COMMIT	Save changes permanently
ROLLBACK	Undo changes

Command	Purpose
SAVEPOINT	Create a checkpoint

Example

```
BEGIN;
UPDATE employee
SET salary=50000
WHERE id=1;
```

```
COMMIT;
```

Easy Memory Trick

 COMMIT = Save

 ROLLBACK = Undo

5. DCL (Data Control Language)

Used to control user permissions.

Command	Purpose
GRANT	Give permission
REVOKE	Remove permission

Example

```
GRANT SELECT
ON employee
TO user1;
```

```
REVOKE SELECT
ON employee
FROM user1;
```

Easy Memory Trick

 DCL = Control Access

SQL Commands Interview Questions & Answers

1. What are the 5 types of SQL commands?

Answer:

- 1. DDL (Data Definition Language)**
 - 2. DML (Data Manipulation Language)**
 - 3. DQL (Data Query Language)**
 - 4. TCL (Transaction Control Language)**
 - 5. DCL (Data Control Language)**
-

2. What is DDL?

Answer:

DDL is used to create, modify, and delete database objects like tables.

Commands:

- CREATE**
 - ALTER**
 - DROP**
 - TRUNCATE**
-

3. What is DML?

Answer:

DML is used to insert, update, and delete data from tables.

Commands:

- INSERT**
 - UPDATE**
 - DELETE**
-

4. What is DQL?

Answer:

DQL is used to retrieve data from the database.

Command:

- **SELECT**
-

5. What is TCL?

Answer:

TCL is used to manage transactions in a database.

Commands:

- **COMMIT**
 - **ROLLBACK**
 - **SAVEPOINT**
-

6. What is DCL?

Answer:

DCL is used to control user permissions and access.

Commands:

- **GRANT**
 - **REVOKE**
-

7. What is CREATE?

Answer:

CREATE is used to create a new database object such as a table.

Example:

```
CREATE TABLE Employee (  
  ID INT,  
  Name VARCHAR(50)  
);
```

8. What is ALTER?

Answer:

ALTER is used to modify the structure of an existing table.

Example:

**ALTER TABLE Employee
ADD Salary INT;**

9. What is DROP?

Answer:

DROP permanently deletes a table from the database.

Example:

DROP TABLE Employee;

10. What is TRUNCATE?

Answer:

TRUNCATE removes all rows from a table but keeps the table structure.

Example:

TRUNCATE TABLE Employee;

11. Difference Between DELETE and TRUNCATE?

DELETE	TRUNCATE
Removes selected rows	Removes all rows
Uses WHERE clause	No WHERE clause
Slower	Faster
Table remains	Table remains

12. What is INSERT?

Answer:

INSERT adds new records to a table.

Example:

```
INSERT INTO Employee  
VALUES (1,'Rahul');
```

13. What is UPDATE?

Answer:

UPDATE modifies existing records.

Example:

```
UPDATE Employee  
SET Name='Priya'  
WHERE ID=1;
```

14. What is DELETE?

Answer:

DELETE removes records from a table.

Example:

```
DELETE FROM Employee  
WHERE ID=1;
```

15. What is SELECT?

Answer:

SELECT is used to retrieve data from one or more tables.

Example:

```
SELECT * FROM Employee;
```

16. What is COMMIT?

Answer:

COMMIT permanently saves changes made during a transaction.

17. What is ROLLBACK?

Answer:

ROLLBACK cancels changes made during a transaction.

18. What is SAVEPOINT?

Answer:

SAVEPOINT creates a checkpoint inside a transaction so you can rollback to that point later.

19. What is GRANT?

Answer:

GRANT gives permissions to a user.

Example:

**GRANT SELECT
ON Employee
TO User1;**

20. What is REVOKE?

Answer:

REVOKE removes permissions from a user.

Example:

**REVOKE SELECT
ON Employee
FROM User1;**

Most Asked Fresher Questions

- ☒ What are the 5 types of SQL commands?
- ☒ Difference between DDL and DML?
- ☒ Difference between DELETE, DROP, and TRUNCATE?
- ☒ What is COMMIT and ROLLBACK?

- ☒ What is **SELECT**?
- ☒ What is **GRANT** and **REVOKE**?
- ☒ What is **DDL**, **DML**, **DQL**, **TCL**, and **DCL**?